Borderless Technical Architecture Plan

1. Architecture Overview

Borderless should be built as a modular trade-orchestration platform with a shared transaction and opportunity core. The architecture should avoid disconnected products and allow later layers to attach to the same trade journey:

```text
Market Access -> Trust -> Match/Negotiate -> Transaction -> Finance -> Compliance -> Logistics -> Payment -> Intelligence
```

Phase I should remain deliberately narrow: Market Access + Trust, with managed sourcing and human operational support. Later capabilities must be extensible without being prematurely built into the MVP.

2. Recommended Tech Stack

Frontend

- Next.js
- React
- Tailwind CSS
- TypeScript

Backend

- Node.js with NestJS or Express
- REST APIs and event-driven modules

Database

- PostgreSQL for transactional data
- Redis for caching and queues

Search

- Typesense or Meilisearch

File Storage

- S3-compatible object storage

Auth

- Auth0, Clerk, or a self-hosted identity provider

Messaging and Notifications

- Email provider
- Optional SMS or WhatsApp integration

Hosting

- Managed cloud infrastructure
- Containerized deployment
- CI/CD pipeline

3. Core Domain Modules

- Auth module
- Company and profile module
- Product catalog module
- RFQ and quote module
- Messaging module
- Opportunity and managed-sourcing module
- Transaction module (Phase II)
- Finance module (Phase IV)
- Logistics module (Phase III)
- Notification module
- Admin module

- Intelligence/AI module (Phase V)

4. Suggested API Domains

- Auth
- Companies
- Products
- RFQs
- Messaging
- Escrow
- Logistics
- Notifications
- Admin
- AI

5. Data Model Principles

- Every transaction entity should have a unique ID
- Sensitive documents should have controlled access
- Escrow events should be auditable
- AI outputs must be reviewable and traceable

6. Security Requirements

- Role-based access control
- Encryption in transit and at rest
- Secure document uploads
- Rate limiting
- Audit logs
- Admin action logging
- Fraud/suspicious-activity reporting
- Privacy and access controls by user, company, opportunity, and document

- Verification and administrative decisions recorded in an audit trail

7. Implementation Phasing

Phase 0: Manual Validation

- Manually facilitate buyer-exporter opportunities.

- Capture requirements, counterparties, documents, logistics needs, financing needs, breakpoints, outcomes, and willingness to pay.

Phase I: Market Access + Trust

- Authentication and role-based access

- Company profiles and verification workflow

- Product catalog and discovery search

- Buyer/supplier/company discovery

- RFQs, supplier responses, and early negotiation

- Managed sourcing and opportunity records

- Messaging and notifications

- Admin moderation and basic event analytics

Phase II: Transaction Management

- Quotes, purchase orders, contracts

- Document management

- Transaction milestones

- Payment coordination

Phase III: Logistics

- Freight, customs, transport, warehousing, insurance, and tracking integrations

Phase IV: Finance

- Purchase-order finance, invoice finance, working capital, trade credit, FX, and insurance partner integrations

Phase V: Intelligence

- Pricing, market intelligence, risk signals, AI sourcing, demand forecasting, and trade analytics

8. Deployment and Operations

- Managed database and object storage
- Monitoring and error tracking
- Backup and recovery
- Incident response process

9. Recommended Phase I Build Order

1. Authentication, roles, and permissions
2. Company profiles and verification
3. Product listings and agricultural discovery
4. RFQs, supplier responses, and comparison
5. Managed sourcing and opportunity workflow
6. Messaging, notifications, and operational notes
7. Admin moderation, audit logs, and event analytics

Escrow, logistics, finance, and AI are intentionally excluded from the Phase I build order and follow the later roadmap phases above.